

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 25

Strong Markov Property

- We have already seen that a MC $\{X_n : n \geq 0\}$ satisfies the Markov property, *i.e.*, for each time m , conditional on $X_m = i$, the process after time m starts afresh from i .
- What if we replace the deterministic time m by a random time?
- For example, if we consider the random time T^i , then what can be said about the process after time T^i ?

Stopping Time

Def: A random variable T taking values in the set $\{0, 1, 2, \dots\} \cup \{\infty\}$ is called a stopping time of a MC $\{X_n : n \geq 0\}$ if the event $\{T = n\}$ depends only on $X_0, X_1, X_2, \dots, X_n$ for all $n = 0, 1, 2, \dots$

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Example 1: For $j \in S$, T^j is a stopping time as

- $\{T^j = 0\} = \{X_0 = j\}$.
- For $n \geq 1$, $\{T^j = n\} = \{X_0 \neq j, \dots, X_{n-1} \neq j, X_n = j\}$.

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Example 2: For $i \in S$, let $L^i = \sup \{n \geq 0 : X_n = i\}$. Then L^i is not, in general, a stopping time as the event $\{L^i = n\}$ depends on whether $\{X_{n+m}\}_{m \geq 1}$ visits i or not.

Strong Markov Property

Theorem: Let $\{X_n : n \geq 0\}$ be a MC with TPM P and some initial distribution and T be a stopping time for $\{X_n : n \geq 0\}$. Then, conditional on $\{T < \infty\}$ and $\{X_T = i\}$, $\{X_{T+n}\}_{n \geq 0}$ is a MC with TPM P and initial distribution δ_i and independent of $X_0, X_1, \dots, X_T$.